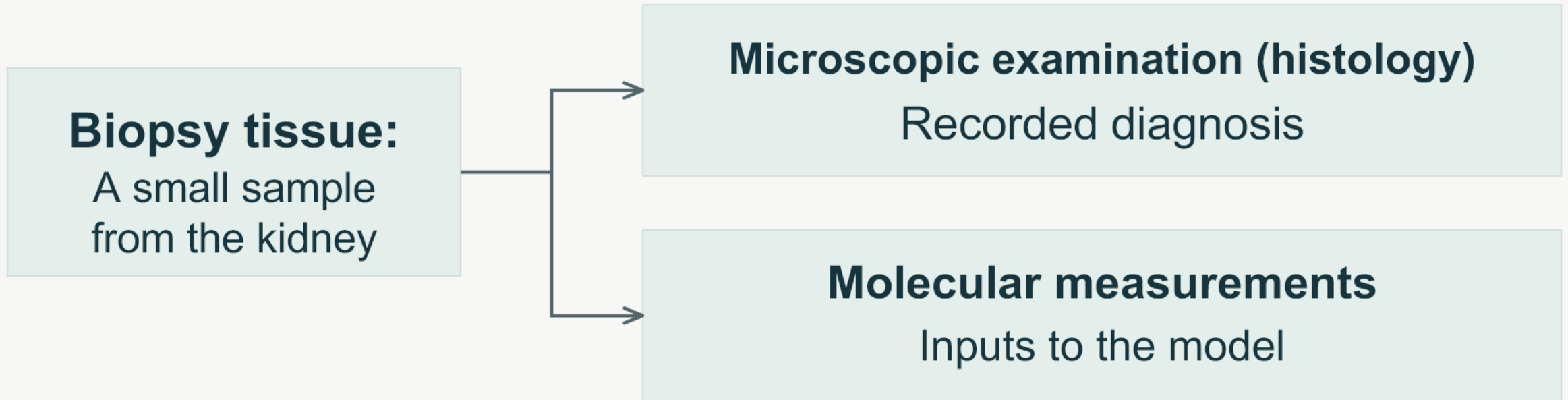


Classifying rejection from kidney biopsy RNA

Can molecular measurements classify
the recorded biopsy diagnosis?

Kidney biopsy and transplant rejection

Rejection: the recipient's immune system attacks the donated kidney.

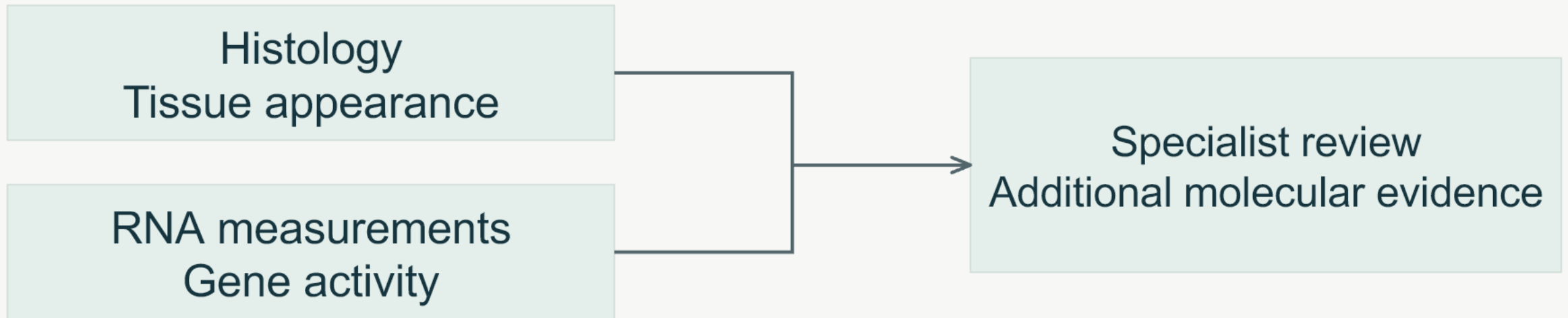


Two kinds of evidence from tissue already collected

Molecular support for uncertain biopsies

Possible use: a molecular second opinion

Borderline findings or disagreement with clinical evidence



This project: learn rejection patterns from recorded diagnoses.

Evidence for a molecular second opinion

Banff guidance

Considers validated molecular tests for difficult biopsy interpretations

B-HOT study Rosales, 2022

Higher initial molecular scores in patients who later developed chronic active antibody-mediated rejection, despite no initial diagnosis of it

RNA can add information beyond the initial tissue diagnosis.

This project tests recorded diagnoses.

Added value in uncertain cases needs direct evaluation.

Recorded diagnoses

Recorded diagnosis	Meaning	Model label
Antibody-mediated rejection	Antibodies: immune proteins	Rejection
T-cell-mediated rejection	T cells: immune cells	Rejection
Mixed rejection	Both rejection processes	Rejection
No rejection	No recorded rejection	No rejection

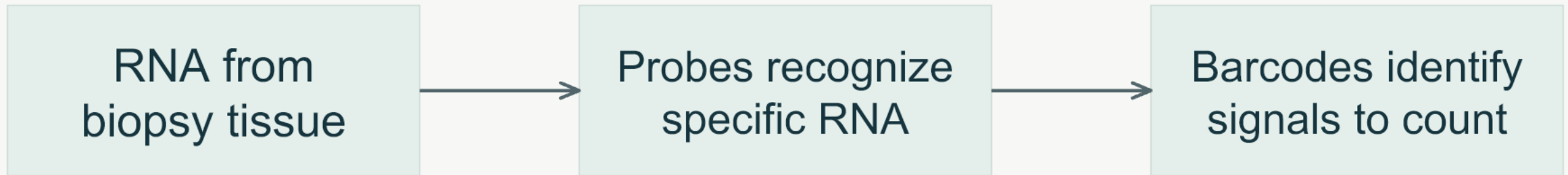
The main question combines all three forms of rejection.

“No rejection” does not mean an otherwise healthy biopsy.

Measuring RNA in biopsy tissue

RNA includes messages cells make when genes are active.

Assay: a laboratory test Panel: a fixed list of molecules to measure



NanoString nCounter, using the B-HOT panel

Example output: IFNG count = 4 in one public specimen

The measurements used by the model

770 RNA measurements per specimen

758

Measurements used as model inputs
Includes some viral RNA signals

12

Housekeeping references
Prepare the other counts

Assay “target”: a molecule a probe recognizes

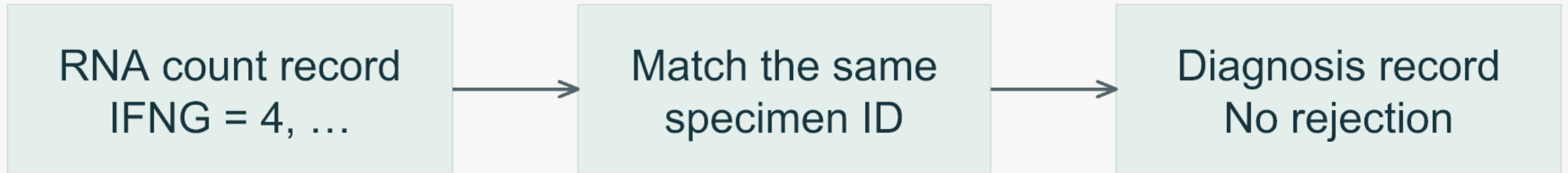
Counts reflect gene activity and the mixture of cells in the tissue.

The 12 reference measurements are removed after normalization.

One row describes one biopsy specimen

1,395 specimens included. No exclusions.

Specimen ID: GSM6510425



Only the prepared RNA measurements enter the model.

IDs, diagnosis and cohort stay separate. No duplicate normalized profiles.

Specimen counts do not establish the number of independent patients.

Preparing counts for the model

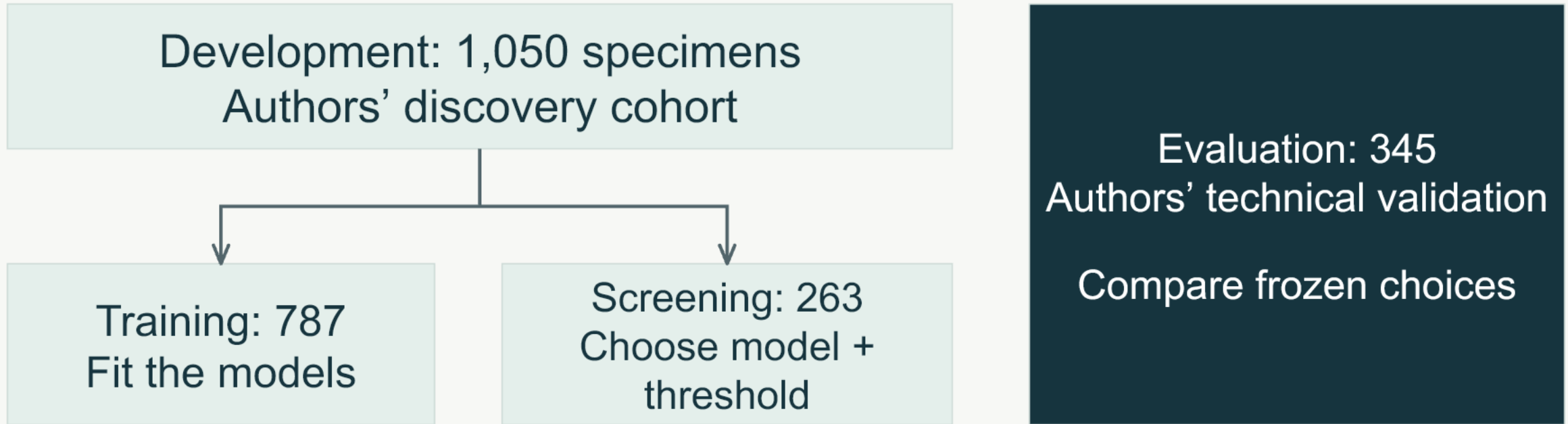
Add 1 to allow zero counts. Take $\log_2$ to compress the range.

IFNG in this specimen	Value
Raw count	4
$\log_2(\text{count} + 1)$	2.322
Mean of 12 housekeeping log counts	7.900
Normalized input: $2.322 - 7.900$	-5.578

Each specimen supplies its own reference. Keep 758 prepared inputs.

Fit any later learned scaling on training specimens only.

Development and evaluation specimens



The saved split keeps the original diagnosis proportions similar.

Patient and center independence are unverified.

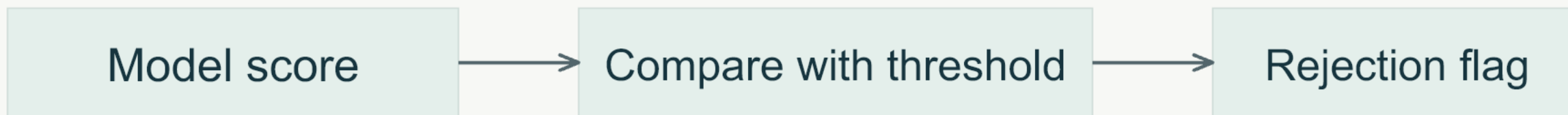
Four model comparisons

Model	What it learns
Constant baseline	One score: the training rejection fraction
IFNG only	A relationship using one RNA measurement
Logistic regression	A weighted combination of 758 measurements
CatBoost	A combination of decision trees

Logistic regularization limits how large the weights can become.

Same specimen preparation. Same evaluation specimens.

Turning a score into a rejection flag



Missed rejection

Recorded rejection, score below threshold

False flag

Recorded no rejection, score at or above threshold

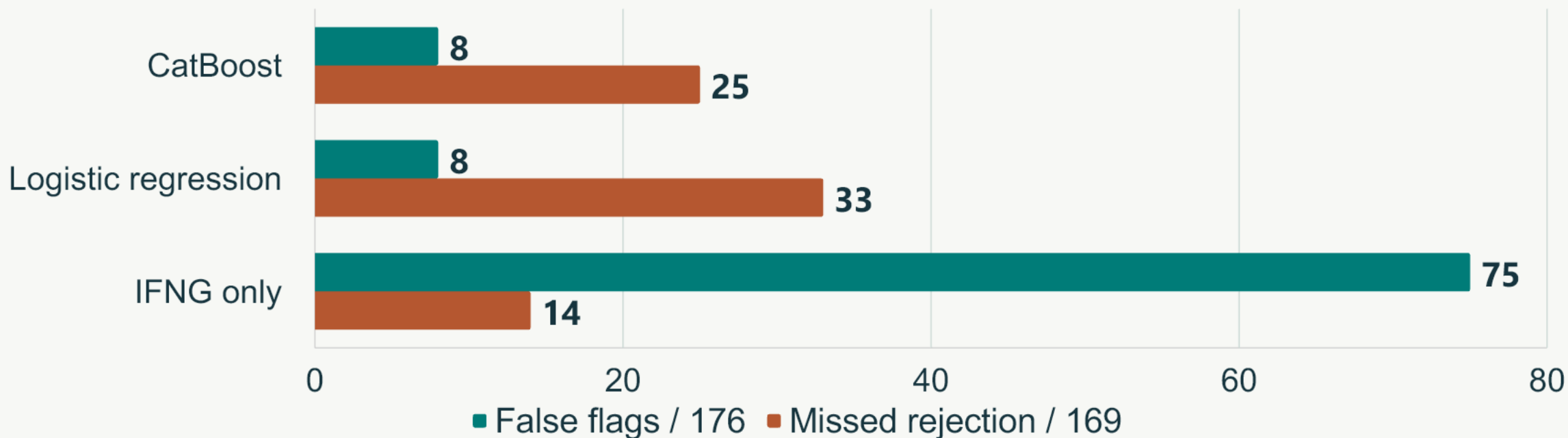
Screening: detect at least 90% of rejection, then minimize false flags.

Freeze the model and threshold together. CatBoost cutoff: 0.876588

90% is an experiment choice. A score is not a calibrated probability.

Several measurements reduced false flags

Same 345 specimens: 169 rejection and 176 no rejection



Constant: 0 / 169 missed, 176 / 176 false flags. It flags everyone.

CatBoost versus IFNG: 67 fewer false flags, 11 more misses.

Most CatBoost misses were T-cell rejection

144 / 169

rejection cases detected

85.2% sensitivity

Screening: 157 / 174 (90.2%)

Evaluation falls below
the 90% selection target

Recorded rejection	Missed / total
Antibody-mediated	6 / 56
T-cell-mediated	18 / 95
Mixed	1 / 18

Errors describe disagreements with recorded diagnoses.

The advantage over logistic regression is uncertain

Fixed models, same evaluation specimens

8 fewer misses among 169 rejection specimens

Both make 8 / 176 false flags. The 95% interval for the detection advantage includes zero (2,000 specimen resamples).

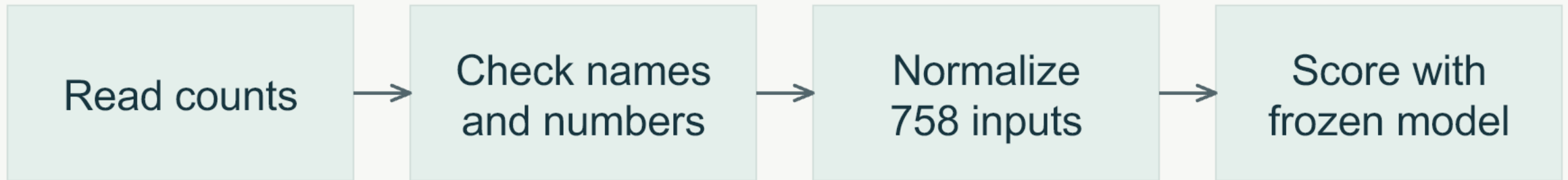
Model choice across 20 development splits

CatBoost	Logistic regression
13 selections	7 selections

The repetitions overlap. Logistic regression remains a simpler alternative.

Training and serving share the same preprocessing

CSV through the command line or web application



Response: specimen ID, score, threshold, flag and model version

Missing or duplicate names, negative or non-finite counts: stop scoring.

Only the configured model can load. Scores remain research scores.

One public specimen through the service

GSM6510425: recorded no rejection

Model score	Frozen threshold	Rejection flag
0.274934	0.876588	False (below cutoff)

IFNG: $\log_2(4 + 1) = 2.322$. Housekeeping reference: 7.900

Normalized input = -5.578

Remove IFNG: “Missing required assay targets: IFNG”

Previous result clears. No new score returns.

Model: 20260915_shared:any_rejection:catboost_all_depth4

The service reproduces all 345 saved scores

345 / 345

API, command line and saved evaluation agree within 10^{-12} .

94 tests passed

Reordered columns: same score. Invalid counts: no score.

Incompatible model metadata: model does not load.

Local research container checked. Cloud deployment remains deferred.

Next evidence for a molecular second opinion

Established: classification in one deposited study

Independence unverified. Assay groups track cohorts.
File checks do not establish laboratory quality.

New cohort

Patient and center IDs, laboratory quality records
Fixed models and thresholds, including logistic regression

Uncertain biopsies

Usual assessment versus assessment plus molecular score
Independent expert review and follow-up outcomes

Test whether the added evidence improves interpretation.

Molecular counts can classify recorded rejection

CatBoost, 345 evaluation specimens

25 / 169 misses. 8 / 176 false flags.

Logistic regression remains a credible simpler alternative.

A tested research prototype for molecular support
of biopsy interpretation

Next: test added value in uncertain biopsies on a new cohort.

Backup: complete primary comparison

Model	Misses	False flags	Sensitivity	Specificity	ROC-AUC
CatBoost	25	8	85.2%	95.5%	0.969
Logistic	33	8	80.5%	95.5%	0.956
IFNG only	14	75	91.7%	57.4%	0.842
Constant	0	176	100.0%	0.0%	0.500

Denominators: 169 rejection; 176 no rejection. Fixed thresholds per model.

Backup: normalization and threshold selection

**Normalized measurement = $\log_2(\text{raw count} + 1)$
– mean of the 12 housekeeping $\log_2(\text{count} + 1)$ values**

Logistic: scaling fitted on training rows; regularization $C = 0.1$.

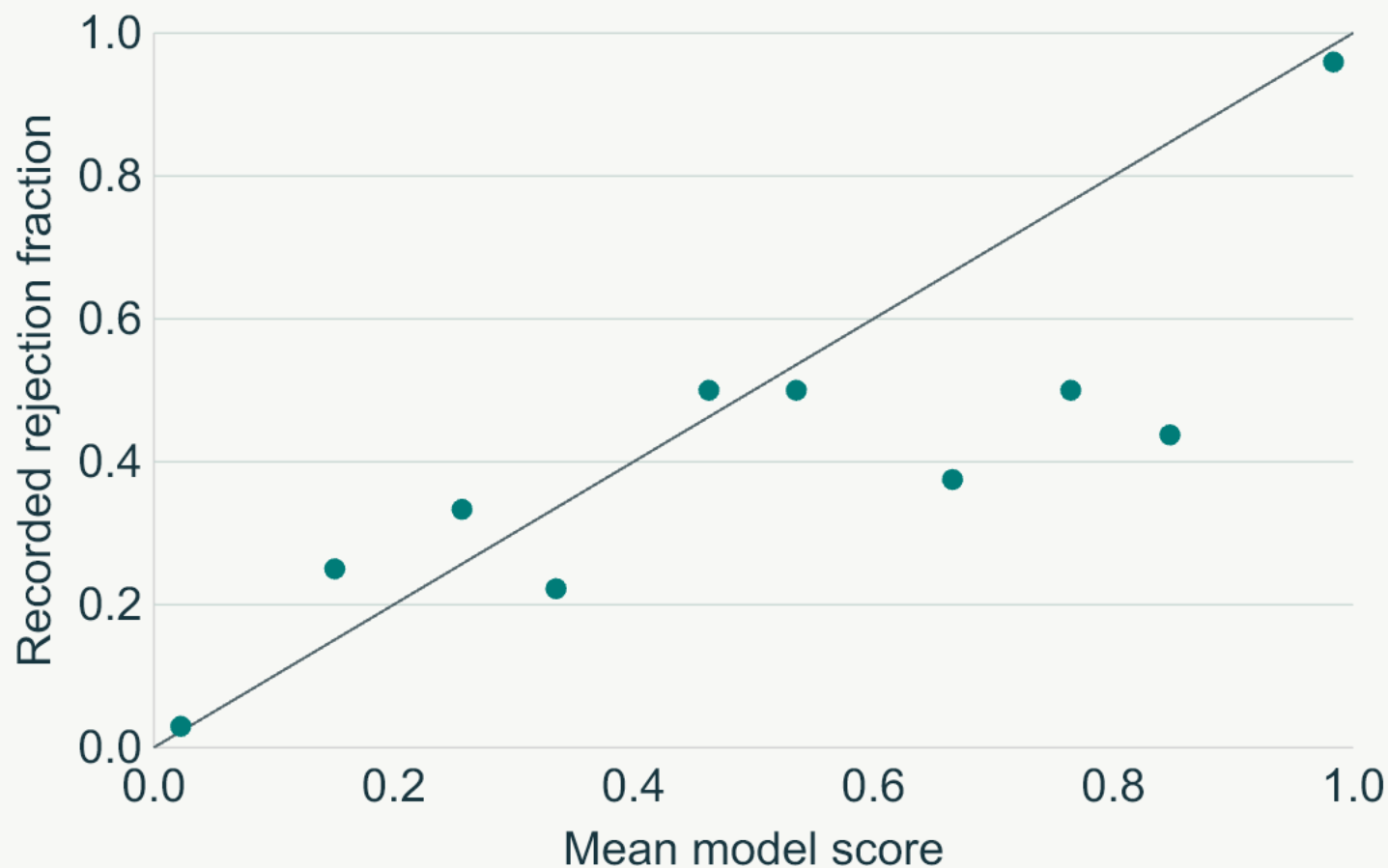
CatBoost: 300 trees, depth 4, learning rate 0.04.

Both detected 157 / 174 screening rejection cases.

False flags: CatBoost 5 / 89, logistic 6 / 89.

Flag when score ≥ 0.8765880870219778 for the frozen service.

Backup: ranking does not guarantee calibrated scores



Score bin 0.8–0.9

0.847

mean score

7 / 16

had recorded rejection
43.8% (95% CI 23.1–66.8%)

Backup: subtype follow-up retained the binary service

Model	Any-rejection misses	False flags
Binary CatBoost	25 / 169	8 / 176
Four-class CatBoost	29 / 169	8 / 176
Four-class logistic	28 / 169	10 / 176
Separate component models	24 / 169	15 / 176

Four-class logistic recognized 10 / 18 mixed diagnoses, with 5 false mixed calls among 327 non-mixed specimens.

Backup: study, assay and deployment limits

Question	What the project establishes
Patient / center independence?	Independence not verified
Laboratory assay quality?	File checks only; lab quality unverified
Individual probabilities?	Calibration remains limited
Cloud deployment?	Local container tested; cloud work deferred

Biopsy-level resampling may understate uncertainty if specimens are related.

Backup: sources and contributions

Public study

Zhang et al., Laboratory Investigation (2024)
GSE212160 raw NanoString B-HOT measurements

This project

Binary comparison and runs produced in this repository,
shared software and a tested local demonstration

Presentation approach

Bourne (2007); Kosslyn et al. (2012); Garner & Alley (2013); Rougier et al. (2014)

Backup: UNOS research connection

UNOS: anatomical issues in donor kidney photographs

Historical labels: both kidneys transplanted,
or an organ refused because of anatomy concerns

Intended use: support clinical assessment and improve consistency

Connection: learn from recorded assessments
to develop additional evidence for specialists

Different clinical task. This is not UNOS validation of our classifier.